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UNIVERSITÉ D'OTTAWA · UNIVERSITY OF OTTAWA

ORBIT NORTH

Canadian Summer Space Intensive — 3 credits

GNG 3100 · SYS 5186

Summer 2026

Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

v3 — 2026-05-05

Canadian Summer Space Intensive — 3 credits GNG 3100 — Topics in Engineering I · SYS 5186 — Advanced Topics in Systems Engineering · Summer 2026

Course Information

Format. 3-credit, **12-day intensive program** delivered in person at the University of Ottawa. The course runs from **Thursday 6 August** to **Friday 21 August 2026** inclusive, on weekdays only — twelve contact days across three calendar weeks. Final dates and any holiday adjustments will be confirmed in Brightspace.

Block	Dates	Days	Hours	Location
Week 1 — Canadian Context, CDF Intro & Concept Development	Thu – Fri 6 – 7 Aug 2026	2	09:00 – 17:00	uOttawa, room TBC
Week 2 — MCR, Regulatory & CDF Intensive	Mon – Fri 10 – 14 Aug 2026	5	09:00 – 17:00	SpaceCDF lab
Week 3 — Operations & Mission Simulation	Mon – Fri 17 – 21 Aug 2026	5	09:00 – 17:00	uOttawa Cyber-range

Mode. In-person. A Zoom link may be available for guest-speaker sessions, but the CDF and simulation components require physical attendance.

Instructor Information

Lead Instructor. Dr Franz Newland · `fnewland@uottawa.ca` · SEDTI, Faculty of Engineering, University of Ottawa.

Industry engagement. Guest speakers from Canadian space organisations. Confirmed speakers will be posted in Brightspace.

Communication. In person before and after class, or by email. Because of the intensive format, please allow 3 business days for replies during teaching weeks. Please read this syllabus in full before emailing a question — many answers are here, and students may be redirected to the syllabus when the answer is already available.

Intended Audience

This course supports differentiated outcomes for undergraduate (GNG 3100) and graduate (SYS 5186) students. It is designed to be interdisciplinary; students from any technical background are welcome and will be deliberately mixed.

Official Course Description

ORBIT NORTH is a 12-day, in-person space systems intensive delivered by the University of Ottawa School of Engineering Design and Teaching Innovation (SEDTI). Students move from the Canadian space ecosystem and its regulatory environment, through a complete preliminary CubeSat mission design produced in a Concurrent Design Facility, to the operator seat in a live mission simulation. The course is designed as a primer for professional practice in the Canadian and international space sector for students with no prior space background.

Additional Course Description

Week 1 (2 days) introduces the Canadian space ecosystem on Thursday morning, then opens the SpaceCDF platform on Thursday afternoon for a short regulatory teaser and the start of mission concept development. Friday is given over to mission architecture and concept work in the CDF, with a first touch on the risk register at end of day. Students are expected to do structured pre-work over the weekend in preparation for the Week 2 Mission Concept Review.

Week 2 (5 days) opens with the Mission Concept Review (MCR) on Monday morning, after which the team's MCR-approved concept is locked. Monday afternoon covers the Canadian regulatory landscape in depth — the Remote Sensing Space Systems Act, spectrum licensing through ISED, export controls, debris mitigation, and registration. The remainder of the week is the SpaceCDF intensive: System-V briefing and the requirements baseline on Tuesday (PRR and SRR are explicitly collapsed into CDF Day 1); power, AOCS, and thermal on Wednesday; comms, structure, and propulsion on Thursday with two formal trade studies; integration, V&V, cost, and risk closure on Friday morning; and a Preliminary Design Review (PDR) on Friday afternoon.

The systems-engineering frameworks themselves — System-V, requirements engineering, functional decomposition, margin policy, trade studies, risk, and planning — are learned **inside** the CDF, because the SpaceCDF platform embeds the SE process scaffolding, the risk register, the planning baseline, and a regulatory-check module as live tools. This shifts the balance from lecture-led foundations to tool-led foundations.

Week 3 (5 days) is dedicated to mission operations and is delivered at the uOttawa Cyberrange. Two days of operations training (ground systems, flight dynamics, mission control system, planning tool, procedures) are followed by two days running a live mission simulation: each team takes the operator seat for a portion of the simulation while another team observes and documents, then the teams swap. The week closes with a structured cohort-internal debrief and final mission review.

There are no required textbooks to purchase. All readings, tools, and software access — including SpaceCDF — will be provided through Brightspace and the uOttawa Cyberrange environment.

Indigenous Affirmation

ANISHINÀBE. Ni manàdjiyànànig Màmìwininì Anishinàbeg, ogog kà nàgadawàbandadjig iyo akì eko weshkad. Ako nongom ega wìkàd kì mìgiwewàdj. Ni manàdjiyànànig kakina Anishinàbeg ondaje kaye ogog kakina eniyagizidjig enigokamigàg Kanadàng eji ondàpinangig endàwàdjin Odàwàng. Ninisidawinawànànig kenawendamòdjig kije kikenindamàwin;

weshkinìgidjig kaye kejeyàdizidjig. Nigijeweninmànànig ogog kà nìgànì sòngideyedjig; weshkad, nongom; kaye àyànikàdj.

ENGLISH. We pay respect to the Algonquin people, who are the traditional guardians of this land. We acknowledge their longstanding relationship with this territory, which remains unceded. We pay respect to all Indigenous people in this region, from all nations across Canada, who call Ottawa home. We acknowledge the traditional knowledge keepers, both young and old. And we honour their courageous leaders: past, present, and future.

Inclusion

The University of Ottawa aims to be an equitable and inclusive institution, actively participating in ensuring the well-being of students, personnel, and faculty members. The University is committed to eliminating obstacles to student inclusion in accordance with the Ontario Human Rights Code. If you would like an alternative name or pronouns used for you during the course, please email the instructor. If you have experienced discrimination or harassment, you can seek confidential assistance through the University Human Rights Office.

The intensive format can raise accessibility questions that a standard 12-week course would not. If health, neurodiversity, caregiving, or financial considerations affect your ability to take part fully, please reach out before the course begins so we can plan together.

Course Learning Outcomes

Learning outcomes are differentiated between the undergraduate (GNG 3100) and graduate (SYS 5186) codes. All students pursue the shared outcomes; graduate students additionally demonstrate the graduate-level outcomes across the same activities.

SHARED OUTCOMES — BOTH CODES

- Describe the Canadian space ecosystem (past, present, emerging) and situate a given mission concept within it.
- Apply the systems-engineering scaffolding embedded in a Concurrent Design Facility to structure a space mission concept through scope, ConOps, functional architecture, and preliminary budgets, drawing on ECSS and the NASA Systems Engineering Handbook.
- Navigate Canadian satellite regulatory obligations — the Remote Sensing Space Systems Act, radiocommunication licensing, export controls, and debris mitigation — at a level sufficient to identify when professional advice is required.
- Contribute substantively to a Concurrent Design Facility intensive design activity, taking a discipline role and delivering a complete preliminary CubeSat mission design across all principal subsystems.

- Maintain a risk register and a planning baseline inside the CDF tool through the full design week.
- Execute spacecraft operations procedures in a mission control environment, including nominal operations, planning, and anomaly response.
- Collaborate in interdisciplinary student teams; give and receive structured technical critique.
- Reflect critically on your own learning trajectory and on the ethical dimensions of space activity.

ADDITIONAL GRADUATE OUTCOMES — SYS 5186 ONLY

- Critically evaluate architecture trade-offs using quantitative methods (margin-based budgets, risk-adjusted comparison) and defend the evaluation in writing and in a review setting.
- Lead a sub-area of the team's mission design and operations, coordinating contributions from other team members and managing interface definition.
- Synthesise lessons learned from the simulation into a structured improvement proposal grounded in the systems engineering and human-factors literature.
- Situate your experience in the course within at least one strand of the scholarly literature on space systems engineering, operations, or new-space practice.

Teaching Philosophy

My role on this course is to accompany you into a professional community. I bring standards, tools, and a network of colleagues — you bring judgement, curiosity, and the willingness to make decisions under uncertainty. The most important thing we will do together is practise thinking about missions as whole systems, and practise making decisions when the engineering is only part of what matters. You will be asked to do things you don't yet feel qualified to do. That is not an accident; it is the core of the course. Many space projects involve doing something for the first time.

Teaching Strategies and Approaches

- Short, concentrated lectures (usually ≤ 45 minutes) anchored to open NASA / ECSS material, used to set up tool-led work.
- A 4-and-a-half-day Concurrent Design Facility intensive in which each student takes a discipline role and the team converges on a complete preliminary mission design in real time, supported by the SpaceCDF platform's embedded SE process, risk, planning, and regulatory-check modules.
- Structured space-industry guest interactions throughout.
- A live, team-based mission simulation at the uOttawa Cyberrange, with alternating operator and observer roles.
- Peer-critique sessions structured around real review gates (Mission Concept Review, Preliminary Design Review).

- Individual reflective writing throughout the course.

A note on review-gate terminology

The course uses NASA terminology: the Mission Concept Review (MCR) is the end-of-Pre-Phase-A gate, and the Preliminary Design Review (PDR) is the end-of-Phase-B gate. ECSS reuses the abbreviation "MCR" for the Mission Close-out Review at the end of Phase E; we do not use that meaning here. The Phase A intermediate reviews (Preliminary Requirements Review, System Requirements Review) are explicitly **collapsed into CDF Day 1** in this course; their content is delivered inside the SpaceCDF requirements module.

Required Materials

All required readings will be provided. Core open references students should bookmark:

- NASA CubeSat 101 — Basic Concepts and Processes for First-Time CubeSat Developers (`https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf`).
- NASA Systems Engineering Handbook (SP-2016-6105 Rev 2) (`https://www.nasa.gov/reference/systems-engineering-handbook/`).
- ECSS-E-ST-10C Rev.1 — System Engineering General Requirements (freely available from ECSS).
- Cal Poly CubeSat Design Specification, Rev 14 (CDS Rev 14).
- SpaceCDF platform — web access and Learner's Workbook provided at the start of the course.
- Optional self-study: the Space Systems Engineering curriculum (Lisa Guerra, NASA, Texas Space Grant) at `https://spacese.spacegrant.org` . The course no longer follows this module-by-module — its content is delivered through the SpaceCDF tool — but it remains a useful reference.

Assessment Strategy

Assessment is portfolio-based. The table below is the default weighting for GNG 3100. SYS 5186 weightings differ as noted. Detailed rubrics will be posted in Brightspace at the start of the course.

Assessment	Type	GNG 3100	SYS 5186	Timing
Pre-course readiness quiz	Individual	5 %	3 %	Day 1, Week 1
Mission Concept Review (MCR)	Team + Individual	15 %	15 %	Day 1, Week 2
CDF daily worksheets and participation	Individual	10 %	5 %	Throughout Week 2
Preliminary Design Review (PDR)	Team + Individual	20 %	22 %	Day 5, Week 2
Operations training quiz	Individual	5 %	2 %	Day 2, Week 3
Simulation performance — operator shift	Team	15 %	18 %	Days 3 – 4, Week 3
Observer critique report	Team + Individual	10 %	10 %	Days 3 – 4, Week 3
Final mission review	Team	10 %	5 %	Day 5, Week 3
Individual reflective portfolio (UG: structured reflection; Grad: critical literature-informed analysis, 2500 – 3000 words)	Individual	10 %	20 %	1 week after Week 3
Total		**100 %**	**100 %**	

Time Commitment

A 3-credit intensive has the same total workload as a standard 3-credit course, compressed into twelve contact days. Expect:

- Approximately 35 hours of scheduled contact per full week (28 hours in the 2-day Week 1).
- 5 – 10 hours of evening preparation per week, particularly during the CDF and simulation weeks.
- **Structured weekend pre-work between Friday 7 Aug and Monday 10 Aug** to prepare for the Mission Concept Review, and again between Monday 10 Aug and Tuesday 11 Aug to prepare for CDF Day 1.

Plan your schedule accordingly. This course is **not** compatible with a full-time job or another intensive course running in parallel.

Language Expectations

The course is delivered in English, with full flexibility to use French in out-of-class discussion, office hours, and in any written work or presentation a student wishes to submit in French. All course materials are provided bilingually; however, the SpaceCDF and operations tools have English interfaces. Communication effectiveness is assessed throughout the course; selecting the appropriate choice of language and register for a given audience is part of that assessment.

Late Assignments and Grade Revision

All assignments should be submitted by their posted due date and time. In the case of illness or other serious situations, contact the instructor as early as possible; university regulations require late submissions due to illness to be supported by a medical certificate. Formal grade revision requests must follow Academic Regulation A-9. Please contact me first for clarification before pursuing a formal appeal.

Participation Guidelines

You can expect your facilitators to:

- Treat you with respect and hear and value your viewpoints.
- Provide constructive feedback and criticism, returned within 1 week of submission.
- Deliver sessions as scheduled; communicate any adjustments through Brightspace and email.
- Respond to inquiries through appropriate channels within 1 business day during teaching weeks.

Learners are expected to take responsibility for their own success:

- Manage your time; the intensive format is unforgiving of fall-behind patterns.
- Come to each day prepared, having completed required readings and any pre-work.
- Participate positively in team activities — you cannot pass this course as a lone operator.
- Demonstrate academic integrity; all generative-AI use must be declared using the AIG attribution framework introduced in Week 1.

Course Calendar

Dates are indicative. Final dates and any adjustments will be confirmed in Brightspace.

WEEK 1 — CANADIAN CONTEXT, CDF INTRO & CONCEPT DEVELOPMENT · OTTAWA · 6 – 7 AUGUST 2026

Day	AM	PM	Deliverables
Thu 6 Aug	**Canadian Space Ecosystem & Course On-boarding.** Past — CSA history, Anik, Alouette, Radarsat, Canadarm. Present — MDA, Telesat, GHGSat, Kepler, NorthStar, university labs and student missions. Emerging — CSA STDP / FAST / CSEP, lunar Gateway, NEOSSat-class missions. Course onboarding, team formation, discipline-role assignment, SpaceCDF account provisioning, tool tour.	**Regulatory Teaser & CDF Intro.** 30-min regulatory awareness teaser; introduction to the SpaceCDF platform and concurrent design method. Teams open a mission file and begin concept work.	Pre-course quiz; team formed; mission area chosen
Fri 7 Aug	**Mission Architecture & Concept Development (CDF).** Mission alternatives trade in the CDF; ConOps top-level v0; concept v0; first parametric budgets.	**Concept Refinement & Risk Register Open.** Continue concept work; open the CDF risk register (preliminary identification only). End-of-week consolidation and weekend pre-work briefing.	Concept v0; ConOps top-level v0; risk register opened

WEEK 2 — MCR, REGULATORY & CDF INTENSIVE · OTTAWA · 10 – 14 AUGUST 2026

Day	AM	PM	Deliverables
Mon 10 Aug	**Mission Concept Review (MCR).** Concept v1; mission alternatives trade outcome; top-level requirements; programmatic envelope; risk register v0.	**Canadian Regulatory Landscape.** RSSSA; ITU coordination; Radiocommunication Act; ISED CPC-2-6-02 spectrum licensing; Canadian Export Control List; Controlled Goods Programme; ITAR overview; debris mitigation (IADC 25-year, FCC 5-year); UN Registration Convention.	**MCR delivered**; regulatory brief (team, 2 pp.); draft ISED application
Tue 11 Aug	**CDF Day 1 — System-V & Requirements (PRR/SRR collapsed).** System-V briefing; tailoring matrix; lifecycle gate map. From mission objectives to SMART requirements; functional decomposition. First parametric budgets (mass, power, data, link). Power, AOCS, thermal commenced in the afternoon.	(continued)	CDF worksheet 1; requirements baseline; updated risk register
Wed 12 Aug	**CDF Day 2 — Power, AOCS, Thermal.** Orbit selection trade. Solar array and battery sizing. Attitude determination and control selection. Thermal environment analysis and thermal-control concepts.	(continued)	CDF worksheet 2; subsystem trade decisions logged
Thu 13 Aug	**CDF Day 3 — Comms, Structure, Propulsion + Trade Studies.** Link-budget calculation; ground-station planning. Frequency band selection and licensing implications via the CDF regulatory-check module. CubeSat structure selection and deployer compatibility. Propulsion trade. Two formal trade studies executed in the CDF trade-study module — link band and propulsion. Equipment selection in the SpaceCDF library; supplier mapping where possible.	(continued)	CDF worksheet 3; equipment selection list; two trade-study reports
Fri 14 Aug	**CDF Day 4 — Integration, V&V, Cost, Risk Closure.** Interface matrix review; clean closure of mechanical, electrical, data, thermal interfaces. V&V matrix assignment by requirement. Environmental test planning. Cost estimation; bill of materials. Risk register closure; planning baseline review.	**Preliminary Design Review (PDR).** SpaceCDF optimiser run; ECSS-aligned document export (MRD, TS, VP); review and action-item capture.	CDF worksheet 4; interface & V&V matrices; cost; **PDR delivered**; ECSS document set

WEEK 3 — OPERATIONS & MISSION SIMULATION · UOTTAWA CYBERRANGE · 17 – 21 AUGUST 2026

Day	Theme	Topics	Deliverables
Mon 17 Aug	Operations Concept Development	Ground segment architecture; antenna and baseband options. Mission Control System (MCS) familiarisation. Operations procedures. Pass planning and scheduling. Telecommand / telemetry definition. Introduction to the Week 3 mission scenarios.	—
Tue 18 Aug	Mission Operations Training	Console positions and responsibilities; shift structure and handover. Communication protocols. Anomaly response procedures. Flight dynamics. Dry-run walkthroughs of LEOP, nominal operations, and contingency scenarios. Team preparation for simulation days.	Ops training quiz
Wed 19 Aug	Mission Simulation Day 1 — LEOP & Commissioning	Full-day simulated LEOP / commissioning. Team A operates a full shift; Team B observes and documents. Real-time anomaly injection by facilitators. Observers complete a structured critique. Post-sim debrief and lessons learned.	Operator team: sim log and anomaly tickets · Observer team: critique notes
Thu 20 Aug	Mission Simulation Day 2 — Nominal Ops & Contingency	Full-day simulated nominal operations and contingency. Teams swap operator / observer roles relative to Wednesday. Post-sim debrief.	Operator team: sim log and anomaly tickets · Observer team: critique notes
Fri 21 Aug	Wrap-Up & Cohort-Internal Final Mission Review	Morning: structured debrief led by student Flight Directors; observer presentations to operators; cross-team lessons-learned synthesis. Afternoon: course synthesis, Canadian space sector pathways, course evaluation, and final mission review (cohort-internal).	Final mission review; observer critique report; **individual reflective portfolio submitted 1 week after Week 3**

Regulation on Bilingualism at the University of Ottawa

Every student has the right to require that a course be given in the language used to describe the course in the course calendar (Regulation on Bilingualism at the University of Ottawa). Except in programs and courses for which language is a requirement, all students have the right to produce their written work and to answer examination questions in the official language of their choice, regardless of the course's language of instruction.

Notice of Collection of Personal Information for Recording of Lesson Capture

Following the Freedom of Information and Protection of Privacy Act in Ontario and University Policy 90, your personal information is collected under the authority of the University of Ottawa Act, 1965. Any Zoom sessions and simulation video recordings will be retained for purposes consistent with the fulfilment of the course learning activities and outcomes, and may include your name, appearance, image, voice, and messages. If you attend remotely and do not wish to be recorded, you may disable your audio and video. If you attend in person and do not wish to be recorded, please contact the instructor during the first week of class to arrange alternatives. The Cyberrange is intended to be used with cameras for team observation. The information collected following this notice will be retained for one year from the end of the semester.

Generative AI and Attribution

Generative AI use is permitted, and in some activities encouraged, subject to the AIG (Assisted by Generative AI) transparency framework introduced in Week 1.

AIG / GIA attribution: Peters (2023), Logos IA-EN, CC BY-NC-SA 4.0 — `https://mpeters.uqo.ca/en/logos-ia-en-peters-2023/`

You will be expected to attach the AIG badge (or its French form GIA) to any deliverable that used generative AI, along with a short notes section describing how it was used. Passing off AI-generated work as your own, without the attribution, is an academic-integrity violation. Refusing to use AI where it is permitted is fine; what is not acceptable is using it without saying so.